1. Sensativity of Learning Rate:

an inappropriately large learning rate can prevent the neural network from convergence.
the training error oscillates drastically around 70% from 6% when we set lr from 0.01 to 10,
proving that even with a sound network architecture, an uncaliberated learning rate will destroy the learning process.

2. The Golden Pairing for Multiclass Classification:

A. modern DL neural networks heavily relies on specific and proven architectural combinations.

B. for example:

⇒ an appropriate batch size ($\neq 1$).

⇒ ReLU function + He Normal function in the hidden layer.

⇒ SoftMass function + Glorot Uniform function in the output layer.

⇒ Categorical Cross Entropy function + Adam function in the compiler.

3. Batch Size Importance:

scaling up the mini batch size from 1 to 64 shifts the computation from serial to parallel implementation.
this tweak not only push the training error down to 4% but also maximizes GPU throughput by the larger scale of matrix multiplication.